

- 1 (a) (i) When bodies in a system interact, the total momentum of the system remains constant,  
provided no net external force acts on it.

- (ii) Relative speed of approach = relative speed of separation

$$u - (-u) = v_B - v_A$$

$$2u = v_B - v_A \quad \dots\dots\dots (1)$$

Conservation of momentum

$$m_A u - m_B u = m_A v_A + m_B v_B \quad \dots\dots\dots (2)$$

$$m_A \times (1) + (2): (2 m_A u) + (m_A u - m_B u) = (m_A v_B - m_A v_A) + (m_A v_A + m_B v_B)$$

$$3 m_A u - m_B u = m_A v_B + m_B v_B$$

$$v_B = \frac{3m_A - m_B}{m_A + m_B} u$$

(b) (i) 
$$v_{\text{tennis}} = \frac{3m_A - m_B}{m_A + m_B} u = \frac{3(0.62) - (0.059)}{0.62 + 0.059} (4.4)$$
  

$$= 11.7 \text{ m s}^{-1}$$

- (ii) The impact force is much greater than the weight of the tennis ball / basketball.  
(Hence, gravitational force, which is an external force, is neglected when applying conservation of momentum.)

- (iii) Basketball is much greater in mass as compared to the ball, hence  $m_B$  is negligible compared to  $m_A$ .

Using (a)(ii),

$$v_B \approx \frac{3m_A}{m_A} u$$

$$= 3u$$

$$= 3(4.4)$$

$$= 13.2 \text{ m s}^{-1}$$

- 2 (a) The moment of a force about a point is defined as the product of the force and the